

WASP-12b

R-1925

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151228 · created 2026-08-06T01:09:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457384.86863 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.016
Transit depth (Rp/Rs)^2	2.1 +/- 0.46 [%]
Orbital Inclination (inc)	76.81 +/- 3.01
Airmass coefficient 1 (a1)	1.046 +/- 0.017
Airmass coefficient 2 (a2)	-0.041 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.66 %
Best Comparison Star	#3 - [209, 269]
Optimal Aperture	3.29
Optimal Annulus	12.5
Transit Duration (day)	0.105 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.016 vs 0.1178 (deviation 1.45σ)
Duration fit vs published	0.105 d vs 0.125 d (deviation 1.06σ)
Mid-time O–C	0.1 min
Depth SNR	7.7
Residual scatter	1.66 %

Light curve

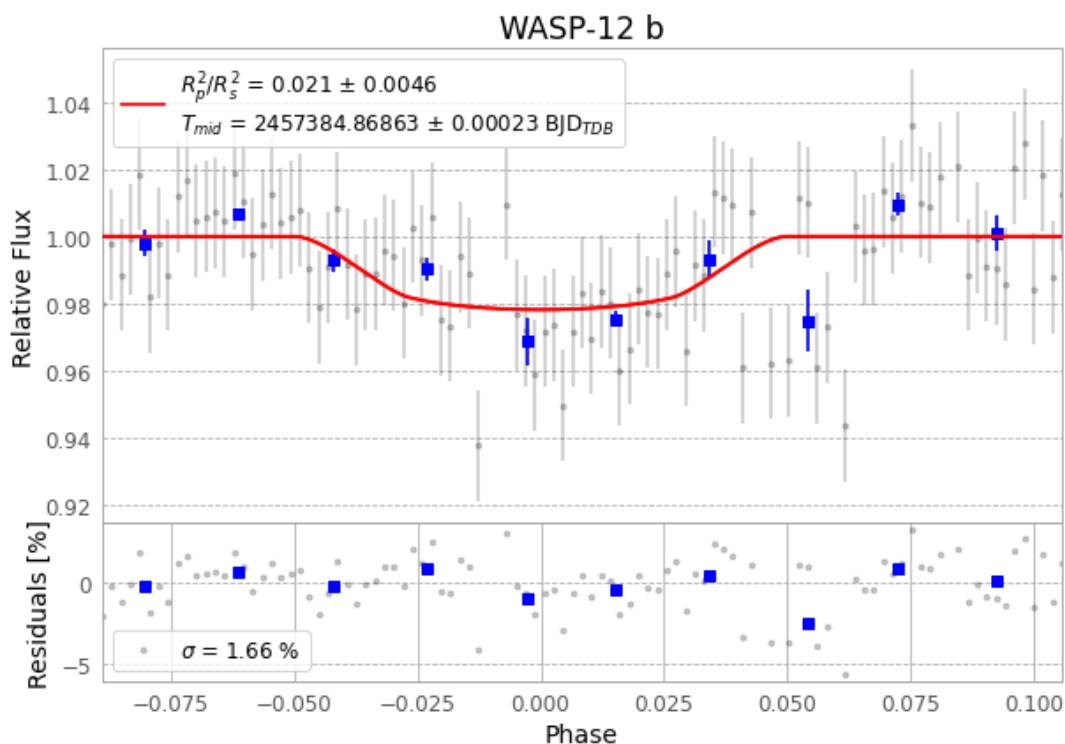


Figure 1 — FinalLightCurve_WASP-12 b_2015-12-27.png (SHA-256 2eca38a17014d6cd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [288, 232], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f914099c7cb3bccc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), WASP-12151228062112.FITS → WASP-12151228112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151228.log (aa1b688d0fbb...), FinalLightCurve_WASP-12 b_2015-12-27.png (2eca38a17014...), FinalLightCurve_WASP-12 b_2015-12-27.csv (2a70a1e20230...)

Compute resources

Wall time 137.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 01:09Z.